Non-linear absorption of intense bi-color laser light by indirect band gap materials resolved by time-dependent density functional theory

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Time-Dependent Density Functional Theory (TD-DFT) is used to generate the transition probability of electrons from valence band (VB) to conduction band (CB) in indirect band-gap semiconductor Si. Two orientations of crystalline lattice are compared. Typical pulses used in laser applications are built using a space of 7 tuning parameters, including wavelength, pulse duration, phases inside the envelope, and the polarization of laser fields. Purpose of this document is for now to analyze and keep a trace of first quantum simulations. This is aimed to prepare a paper on the non-linear absorption of light by Si atoms exposed to single-color and bi-color laser excitations.

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I. INTRODUCTION

It was experimentally demonstrated that a dense electron-hole plasma is generated upon femtosecond laser irradiation[1]. Such transient excitation leads to the change of refractive index as a function of time and laser energy and can lead to excitation of several processes such as triggering chemical reaction, non-thermal phase transitions, and surface plasmon polariton excitation[2,3]. Making use of multi-photonic absorption also enabled outstanding applications such as 2-photon polymerization [REF], and multiphoton microscopy [REF].

First theoretical description of non-linear absorption in matter was successful but are nowadays not sufficient to address the needs for describing laser-induced band gap energy levels, non-parabolic band structure, or the wide spectrum of ultrashort laser pulses. Nowadays, strong hypothesis such as monochromaticity of waves, parabolic band gap approximation, and direct interband transitions[4,5] could not be easily removed. Therefore, calculating the non-linear excitation transition rates in matter remain a puzzling problem.

In addition, making use of several laser pulses together lead to give more control over laser ablation , nanostructure formation[6–9], high harmonic generation and [see results from Les Houches workshop].

In view of bridging large scale simulations[10–12] with first principle calculations[13,14], this paper propose a quantum simulation of the non-linear absorption in indirect band gap material with the example of silicon, based on the method of TD-DFT.

II. CALCULATION METHOD

In the quantum modeling world, the linear absorption in bulk is typically resolved using the meta-Generalized Gradient Approximation (MGGA) qualitatively well reproducing the experimentally measured absorption spectrum for Si and Ge[15]. A full picture of the photo-emission from atoms and molecules also came to be available[16].

Hence, it appears that linear absorption can be calculated using methods like MGGA, or Kohn-Sham approach with exchange-correlation derivated from Bethe-Salpeter equation.

Difficulty appears to be linked with the description of excitons bounded on large computational cell, and with non-linear excitation (intense fields). So far, the case of gas phase ionization was well studied (so called “above threshold ionization”) [4,1718,19].

Today the difficulty remains to calculate excitation rates in correlated matter upon intense laser light. Such calculation started for small metallic systems[20–24]. In addition, the effect of intense constant field got recently

studied[25].

In this paper, we will show results of quantum simulations made on Si, from the most basic case to the most advanced. This will support the development of my work in MPSD. We will select the interesting ones before publication.

A. Preparation of the laser pulses

1. Lasers in quantum simulations

As the scale of the sample is of few atomic layers, the local electric field of the laser can be approximated by its simple temporal variation. Simulations are resolved in a space-time-dependent approach, hence the dispersion of the laser pulse is natively taken into account. For short pulse durations, the spectrum of the pulses becomes wider.

If using QED-DFT[13], some modulations of the electric field response may be observed.

2. Temporal evolution of bi-color pulses

Experimental techniques In the experiments, two configurations are possible.

(i): We can make use of one laser source, and place a BBO crystal in the optical path, then focusing both pulses using the same lens, in the most ideal case. In this case, the two pulses will have blocked relative phase $CEP_2 - CEP_1$.

(ii): Second possibility is to use two different lasers, which may then have two different phases CEP_1 and CEP_2, even maybe time-varying relative phases.

We now draw a general model which takes into account all possible configurations. At the atomic scale, the spatial variation of the pulse envelope can be neglected. Hence, the temporal envelope of the pulse can be defined by the general bi-color pulse expression

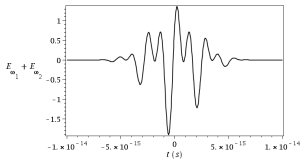
$$\boldsymbol{E}(t) = \boldsymbol{E}_{\lambda_1}(t) + \boldsymbol{E}_{\lambda_2}(t, \Delta t).$$

Each pulse can be developed in

$$\boldsymbol{E}_{\lambda_1} = \boldsymbol{E}_0^{(1)} \sin(\omega_1 t + CEP_1) \cdot \exp\left[-\frac{1}{2}\left(\frac{t}{\sigma(\tau_1)}\right)^2\right]$$

$$\boldsymbol{E}_{\lambda_2} = \boldsymbol{E}_0^{(2)} \sin(\omega_2 t + CEP_2) \cdot \exp\left[-\frac{1}{2}\left(\frac{t-\Delta t}{\sigma(\tau_2)}\right)^2\right]$$

These two pulses defines a space with 8 dimensions to be investigated: $E_0^{(1)}$, $E_0^{(2)}$, ω_1, ω_2, CEP_1, CEP_2 and Δt, but also the orientation P_1 and P_2 of the electric field in real space (polarization) parallel to surface (s or p) state for each pulse.



(a) (b)

Figure 1: Electric field of the bicolor irradiation. (a) Source beams. (b) Mixed beams (800 nm and 400 nm).

Effect of the second wavelength In particular, it is predicted that effect of the relative CEP for the interfered beams will strongly influence the bicolor transition probability. Fig. 1 shows how two beams of first and second harmonic are mixing. It is rather clear that a non-periodic modulation of the electric field will have dramatic consequences on the electron dynamics, hence changing the absorption probability, in particular the nonlinear absorption described by Keldysh[4].

Effect of the pulse phase (CEP) We can also consider two supplementary parameters with adding the pulse durations τ_1 and τ_2. These later parameters are expected to average or increase the importance of CEP parameters. Fig. 2 shows the effect of changing the phase of the second pulse. Pulse becomes asymmetric in respect of field envelope, which then could be considered as harmonic in the non-linear regime, i.e., absorption probability is expected to be also phase depedent.

Effect of the pulse delay Δt Finally, chirped pulses should be considered, but then may require another formulation. Fig. 3 shows the effect of the delay between pulses.

Chirped pulses?

B. Preparation of the sample

We generate a silicon slab with using periodic boundary conditions along the surface, and absorbing boundary conditions in the direction normal to the surface.

Which orientation to choose ? In the context of nanostructure formation, the effects of crystal orientation were only slightly investigated.[26–28] It was shown that crystal orientation should not play a role in the periodic nanostructures of electrodynamic origin. However, it may play a role in the formation thermodynamically formed periodic structures. As Si has a diamond-cubic crystalline structure, we choose Si(100), and Si(111).

2d-periodic Si slab In view of studying the effect of the surface, we build a 2D-periodic Si lattice with periodic boundary conditions in directions X and Y. Fig. 4 shows a primary cell of the Si slab. To avoid inducing metallic behavior at the interface, H terminations were added at the surface. The natural oxidation of Si is here neglected. As samples are considered at ambient temperature, the most probable surface reconstruction is the

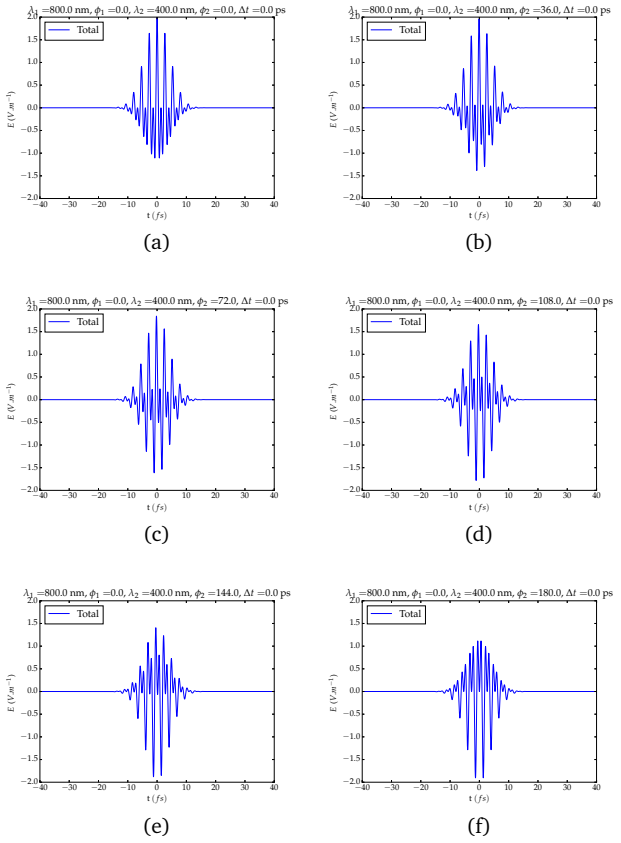


Figure 2: Electric field of the bicolor irradiation. Mixed beams (800 nm and 400 nm) of various respective phases

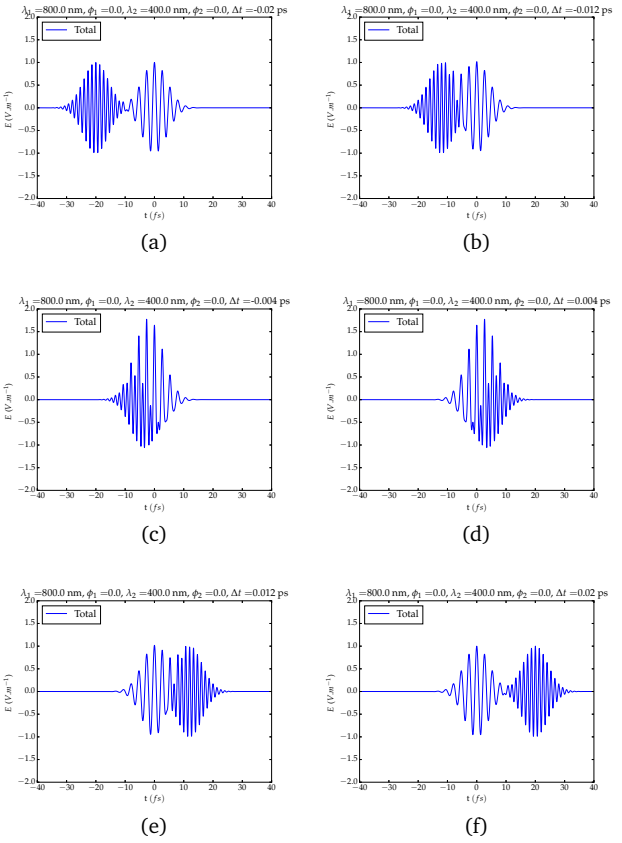


Figure 3: Electric field of the bicolor irradiation. Mixed beams (800 nm and 400 nm) of various respective phases

dihydridic 1×1 occupying the pending bonds of Si[29]. Fig. 1(c-d) show the Si sample with such H terminations.

As we would like to observe the evolution of the wavefunction density $|\psi|^2$ in the sample, we generate two types of samples. Slab 1 is the smallest sample where a large grid of k-points allows for a good convergence of electronic level calculation. Slab 2 is a second slab describing the same material, where the number of atoms is increased in view of maximizing the spatially resolved information. Such procedure does not affect the integrity of calculation, as the whole physical information is contained into the Kohn-Sham wavefunction densities $|\psi|^2$. The size of Slab 2 was increased until exhibiting the same energy per atom as Slab 1. As soon as the sample was thick enough, overlapping between surface induced energy levels and bulk energy levels was suppressed, leading to convergence of the sample to the same total energy.

1. Procedure for building a minimal slab (SLAB 1)

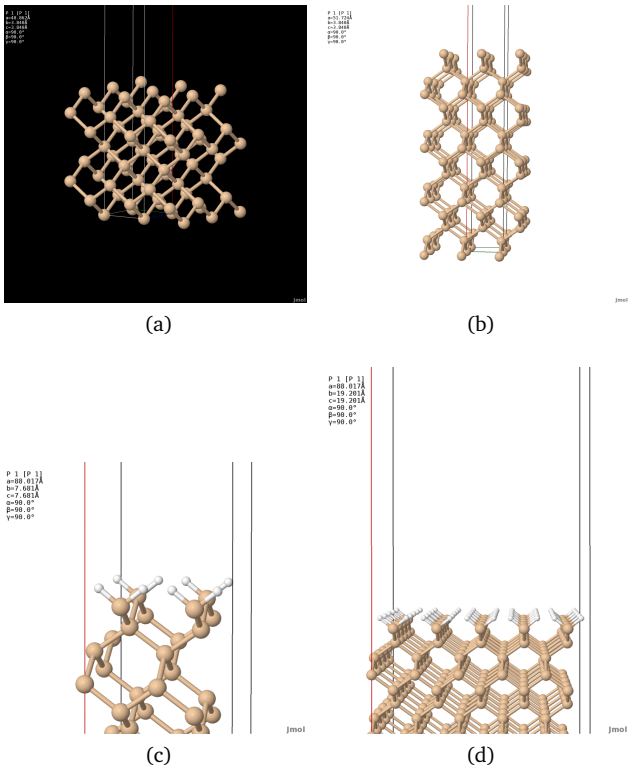
1.: Calculate the energy per atom, via making a bulk calculation. $E_{at} = \frac{E_{tot}}{N_{at}}$

2.: Increase slab depth until convergence of surface energy (see Eq. (2) in paper Beschtedt). Means that both surface are not interacting (wavefunctions are not overlapping any longer).

3.: Once this slab (minimal number of atoms in the plane) is built, we start calculating the k-points to calculate band structure. Find the required number of k points so that electronic total energy per atom does not change.

4.: Once it is obtained, we found the N_e of the ground state ($T = 0$ K). The whole observables are available from this knowledge (DFT).

Fig. 5 presents the result of convergence for a vacuum size $L = 3$ nm. It appears that a k-grid $1 \times 8 \times 8$ is sufficient to reach convergence of the total energy per atom. However, it is unclear if the number of layers is sufficient. "Tier1" and "Tier2" are increasing levels of precision of the function basis of the Si orbitals. As we would like to improve performances to the maximum, we shall reduce the precision to get the biggest box possible.



(a) (b) (c) (d)

Figure 4: Atomic representation of the modeled slab samples. (a) 2 layers with 2 atoms of Si each. (b) 4 layers composed of 2 Si atoms each. The number of atomic layers is increased until convergence of the total energy per atom. (c) Di-hydridized Si(100) noted $Si(100) : H_2$. (d) Same as (c) with more atoms.

2. Building a large SLAB 2

1.: From SLAB1 > Step 2, we have a column which should be repeated in X and Y, using only 1 k-point.

2.: We add atoms in X and Y directions (same number in both directions) until total energy is converged, and equal to SLAB 1 > Step 4 energy.

3.: Then we know that we have built the proper ground state for the supercell of Si.

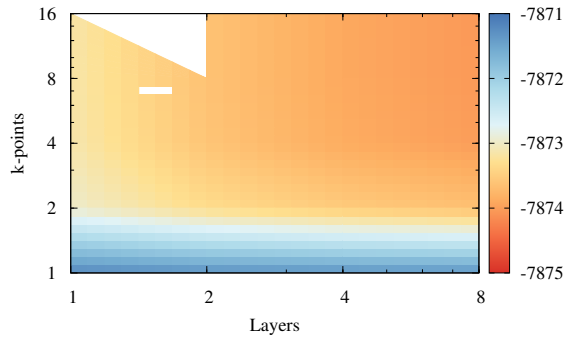
III. SIMULATION RESULTS

We are basically interested in accessing to the temporal density evolution of electrons *in the state* $\boldsymbol{k}$

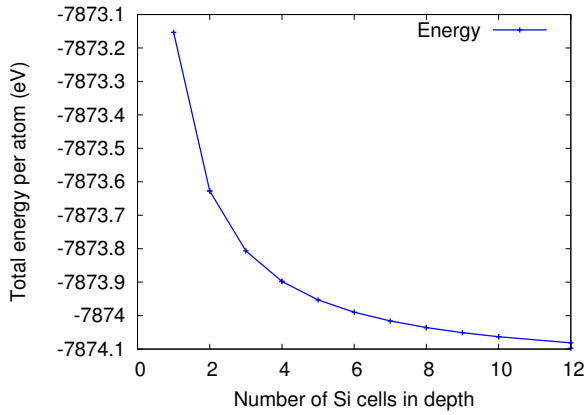
$$\frac{\partial N_{\boldsymbol{k}}}{\partial t}$$

promoted from the valence bands to the conduction band of Si as described by

$$N_{\boldsymbol{k}}(t) = \int \psi_k^{\dagger}(\boldsymbol{r},t)\,\psi_k(\boldsymbol{r},t)\,d\boldsymbol{r}.$$



(a)



(b)

Figure 5: (a) Convergence of total energy as function of Si layers and number of k-points. Each layer contains equivalent of 2 atoms. The number of atomic layers is increased until convergence of the total energy per atom.

As shown recently [REF quantum simulation of linear excitation], the linear processes are exciting electrons towards many wave-vectors of the conduction band. The difficulty is to restrict the number of carriers to the conduction band of Si, which may also evolve in time. In particular, at high laser intensity, ps-melting[30] and non-thermal melting[31,32] can occur, where the band gap undergoes a collapsing effect.

HOW TO SOLVE THIS ?

A. Band structure of Si slab

Show the change of band structure as function of number of layers.

B. Excitation of free-carriers to conduction band with single-color laser irradiation

C. Transition probability of electrons to conduction band upon bi-color laser irradiation

D. Quantum description of the band gap collapse with increasing laser fluence

IV. CONCLUSION

ACKNOWLEDGEMENTS

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